Fiscal Year:	FY 2018	**Task Last Updated:**	FY 05/17/2019
PI Name:	Ryder, Jeffrey Ph.D.		
Project Title:	Identification of Muscle Fitness Standards for Exploration Mission Tasks		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Houston	**State:**	TX
Zip Code:	77058-3711	**Congressional District:**	36
Comments:	NOTE: Formerly at Universities Space Research Association.		
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	10/21/2015	**End Date:**	09/30/2018
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Norsk, Peter	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 9/30/2018, per PI; end date originally 9/30/2017 and then 2/27/2018 (Ed., 7/11/18) NOTE: End date changed to 2/27/2018, per HRP Element; end date originally 9/30/2017 (Ed., 1/31/17)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	English, Kirk Ph.D. (University of Houston-Clear Lake) Ploutz-Snyder, Lori Ph.D. (University of Michigan)		
Grant/Contract No.:	Directed Research		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Current National Aeronautics and Space Administration (NASA) muscle strength standards are vague and have little or no association with planetary functional performance. In advance of future deep space exploration missions, it is thus essential that we develop muscle strength/power standards that are predictive of the successful performance of mission tasks. Benchmark mission tasks have been approved via the NASA Human Research Program (HRP) and Human Systems Risk Board (HSRB) following a thorough vetting process across a broad cross-section of stakeholders (e.g., physiologists, science management, engineers, flight surgeons, astronauts, mission designers, flight operations personnel). These mission tasks were approved because they are both physically challenging and mission-critical. The research in this Directed Task Proposal will entail the characterization of the physiologic requirements to perform the identified tasks that include two nominal and two off-nominal (i.e., emergent) tasks. We will first measure predictor variables such as maximal strength, power, power endurance, etc., in volunteers similar in age and physiologic capacity to the astronaut corps. Subsequently, subjects will perform the selected critical mission tasks. To evaluate the effects of muscle strength/power on task performance, we will employ our weighted suit model to decrease subjects' relative strength (strength/body weight ratio); subjects will complete each critical mission task with an additional 0, 20, 40, 60, and 80% of their bodyweight. We will measure time to completion for each trial and employ both fractional polynomial regression and spline regression techniques to address our hypothesis that a strength (power)/bodyweight threshold exists, below which performance decrements follow a markedly different and negative slope. Results from these models will enable us to inform NASA of the muscle strength/power thresholds below which performance of mission tasks is impaired. Specific Aim: To determine the strength/power thresholds below which performance of critical mission tasks is significantly degraded. Human subject testing will be completed on the ground in the Exercise Physiology Laboratory at NASA's Johnson Space Center (JSC). Approximately 60 test subjects will be included in the study.
Rationale for HRP Directed Research:	This project is in direct response to both an August 2014 HSRB requirement and the baselined Human Research Program (HRP) Path to Risk Reduction milestone of providing updates to the NASA Fitness For Duty Standards. Specifically, the HSRB asked for an update to the muscle strength/performance standard. The new standards should be tied to fitness for duty for exploration tasks and provide a quantitative index of readiness to perform key exploration tasks. This research effort will leverage upon previously HRP-funded weighted suit investigations that are sub-tasks to the Functional Task Test investigation. The weighted suit approach uses a novel analog for assessing the effects of muscle deconditioning on task performance. We have successfully used this approach to threshold the short FTT tasks (Ryder et al. European Journal of Applied Physiology, 113(4):911-21, 2013). Additionally, we have completed proof-of-concept studies using a new suit to demonstrate that it can be successfully worn while performing longer duration tasks. The current newer weighted suit was designed and fabricated with inputs from Exercise Physiology and Countermeasures Personnel specifically for use during the types of extended duration tasks that are needed to be performed in order to establish muscle performance thresholds for exploration. This new research involving highly vetted mission-like tasks is a follow-on to recent HRP-funded efforts and importantly requires specific expertise, technologies, and data analytics that have developed within the Exercise Physiology and Countermeasures Laboratory as part of those earlier efforts. Therefore the work reflects "focused and constrained data gathering and analysis that is more appropriately obtained through a non-competitive proposal". Thus, the proposed works qualifies for directed research under the "Highly Constrained Research" category in the HRP Unique Processes, Criteria, and Guidelines (UPCG).
Research Impact/Earth Benefits:	Results from this work will inform updates to muscle strength standards for space exploration crew in NASA STD-3001 Volume 1. Results may also provide information for establishment of fitness for duty standards for terrestrial occupations such as firefighters. The methodology described by this work will also provide for occupation-specific fitness requirement evaluations for terrestrial occupations.
Task Progress:	Baseline muscle performance measures and exploration task testing were completed on 60 subjects. Subjects performed each of the four exploration tasks under five different loaded conditions. The relationship between strength/weight versus task performance time was characterized. Strength thresholds below which task completion time (i.e., task performance) became markedly impaired were identified. Details of the methodology and results are captured in peer-reviewed article listed in the Bibliography section below.
Bibliography Type:	Description: (Last Updated: 08/11/2025)
Articles in Peer-reviewed Journals	Ryder JW, Fullmer P, Buxton RE, Crowell JB, Goetchius E, Bekdash O, DeWitt JK, Hwang EY, Feiveson A, English KL, Ploutz-Snyder LL. "A novel approach for establishing fitness standards for occupational task performance." Eur J Appl Physiol. 2019 Jul;119(7):1633-48. https://doi.org/10.1007/s00421-019-04152-3 ; PubMed PMID: 31069517 , Jul-2019